

What are the benefits of using GaN instead of silicon in power adaptors?

Power adaptors used to vary for each device due to different connectors. With standardised connectors, chargers are unified. Gallium Nitride (GaN) technology operates at high frequencies and reduces adaptor size by minimising electromagnetic components. GaN adaptors are smaller, more powerful, and more efficient, offering up to 95% efficiency compared to 80-85% in silicon ones. This leads to faster charging speeds and the ability to charge multiple devices with one compact adaptor.

What are the challenges in GaN design?

Significant challenges arise at the PCB level, particularly with noise-complaint routing and external noise prevention. As electronic devices become smaller, selecting the right components to meet size targets is critical. The PCB layout, component selection, and manufacturing must be meticulously planned. Issues such as parasitic inductance and thermal management are crucial, especially when using GaN, which remains cooler than traditional silicon and enhances thermal dissipation. Effective thermal management is essential in high-power applications, requiring expert design of high-frequency adaptors or converters.

How do you combine GaN devices with components like power adaptors?

GaN technology is key in power converters, especially for consumer products like chargers and potentially portable inverters. We focus on integrating GaN into small-scale applications under 1kVA, moving beyond traditional silicon-based systems. This shift is part of a broader trend towards GaN use in smaller electronic devices, from chargers to home power systems,

reflecting its growing standardisation in the industry.

How do you justify the higher cost of GaN devices in product design and marketing?

When designing a product, having control over the entire design IP is crucial. GaN devices may be 10-15% more expensive than silicon, but with your own design, you can optimise costs by adjusting other components. As GaN technology becomes more

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widely adopted, prices will naturally decrease, similar to what happened with BLDC motors used in fans. The key advantage of owning your design IP is the ability to balance cost and performance, making your product more competitive compared to pre-assembled alternatives, like those from China. This control allows for better optimisation and cost management in the final product.

What challenges did you face in scaling GaN technology for mass production, and how did you overcome them?

Scaling GaN technology involves educating consumers on its benefits, such as charging multiple devices with a single, compact charger. Many people were unaware that one

GaN charger with USB-C ports could handle phones, laptops, and wearables. We have run extensive campaigns to overcome hesitancy about using non-branded chargers, emphasising their interoperability, and promoting the idea of one charger for all devices. Our strategy focuses on raising awareness and offering a comprehensive charging solution.

How will GaN technology shape the future of mobile and device chargers?

GaN technology is set to transform the charger market. The global GaN charger market was \$1.1 billion in 2023 and is expected to double by 2024, with a 19.9% annual growth rate thereafter. India, a major consumer electronics market, is seeing high demand for GaN chargers, projected to grow by 18% annually. The Indian government's reduction of customs duties on electronics is boosting local production and exports. With strong demand for chargers, especially in growing markets like India, GaN technology is poised for significant growth.

What emerging trends are there in GaN, and how should companies prepare to integrate them?

GaN technology has evolved from powering satellites with solar panels to being used in electric vehicles, and now, consumer electronics. Initially used in space for lightweight, efficient power converters, GaN is now key in automotive and consumer markets. Research, particularly in Europe and the US, is pushing the boundaries of GaN, aiming for higher power densities and efficiency. With growing adoption in consumer electronics, there is a need to bridge the gap by offering advanced, smart charging solutions. As consumer adoption increases, costs will decrease, and demand will grow, making GaN a significant player across various industries. **EFY**